23.10.2026

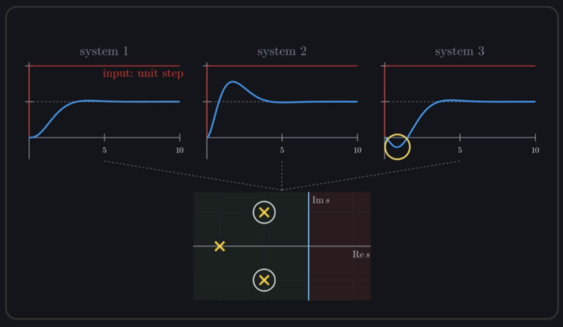
Exercise 6



Goal Today: Poles and zeros, s-Plane

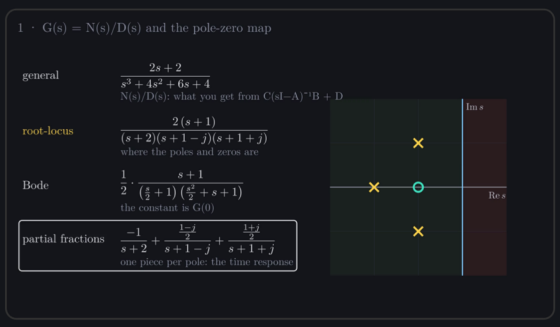
Video 0 · The puzzle

Three systems, the same unit step as input. All three have poles at -2 and $-1 \pm j$. Yet one rises smoothly, one overshoots, and one first moves the **wrong way**.



Video 1 · G(s) = N(s)/D(s) and the pole-zero map

$(sI - A)^{-1}$ = adj/det, so $G(s) = N(s)/D(s)$ with $D(s) = \det(sI - A)$. Poles = roots of D = eigenvalues of A, zeros = roots of N. Factor: every $(s - p)$ is a point in the s-plane. Same function, four ways of writing it.



Poles: eigenvalues of A. They only depend on physics.

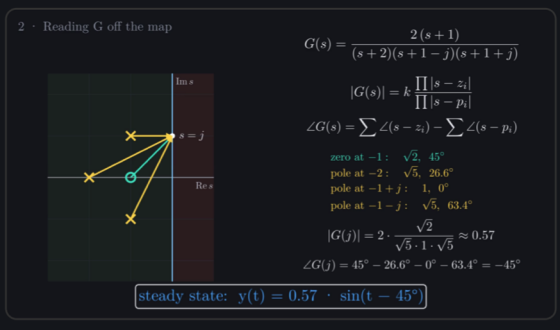
Zeros: also depend on B, C.

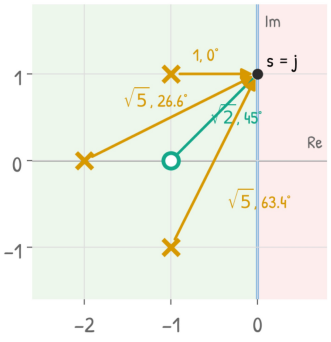
Video 2 · Reading G off the map

Each factor $s - p$ is the **arrow from p to s**. Multiplying complex numbers multiplies lengths and adds angles:

$|G(s)| = k \cdot \Pi|s - z_i| / \Pi|s - p_i| \quad \angle G(s) = \Sigma\angle(s - z_i) - \Sigma\angle(s - p_i)$

At $s = 0$: $G(0)$ = final value of the step response (stable systems).





$|G(j)| \approx 0.57$

$\angle G(j) \approx -45°$

$\sin t \longrightarrow 0.57 \sin(t - 45°)$

- s close to a pole: $|G|$ large
- s close to a zero: $|G| \approx 0$

Made with Goodnotes

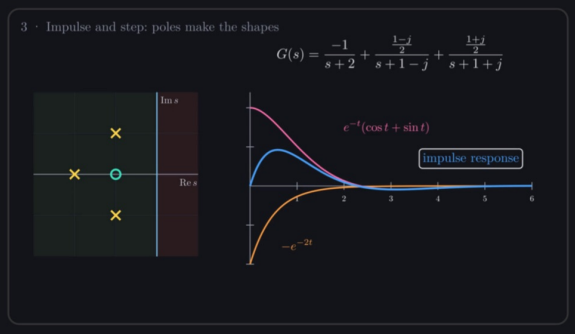
Video 3 · Impulse and step: poles make the shapes

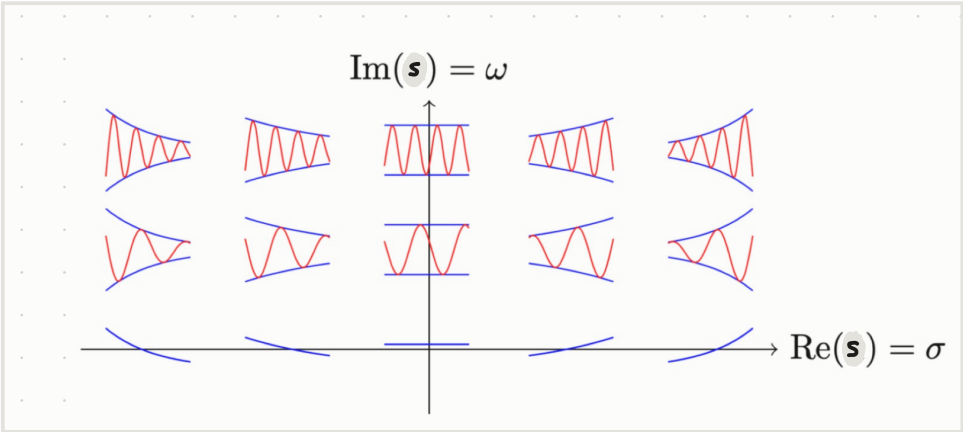
Impulse = one hammer blow (area 1, no duration); step = a constant push held forever. An impulse is an **initial condition in disguise**: it sets $x(0) = B$, so $h(t) = Ce^{At}B$. Every pole contributes one mode:

$G(s) = \Sigma\, r_i/(s - p_i) \implies h(t) = \Sigma\, r_i\, e^{p_i t}$

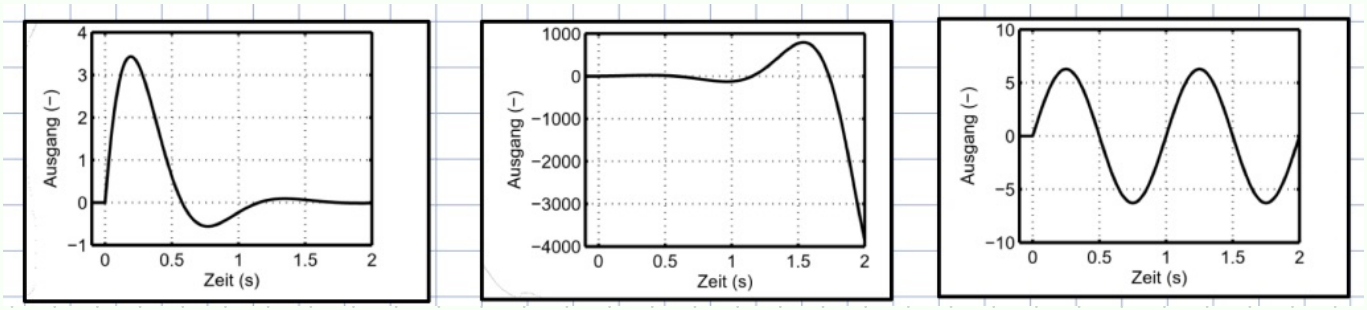
The step response is the running area under h: $y_{step}(t) = G(0) + \Sigma\, (r_i/p_i)\, e^{p_i t}$

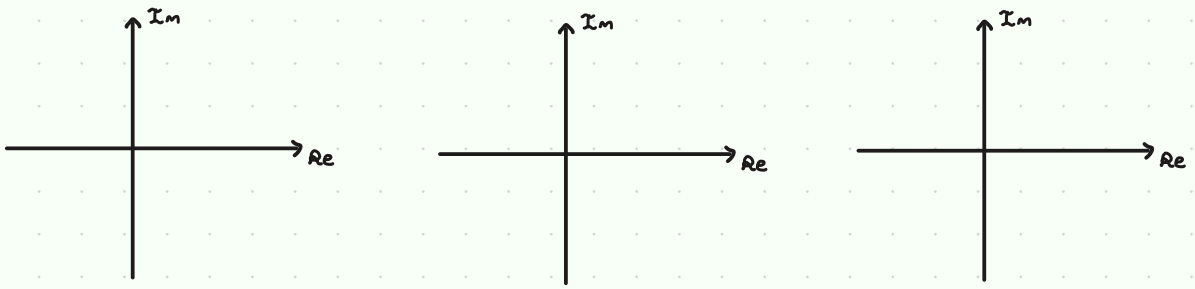
Initial and final value theorems: start at $s \to \infty$, end at $s = 0$ (step: G, impulse: s·G).





Your turn: Draw the poles of the systems by their impulse response



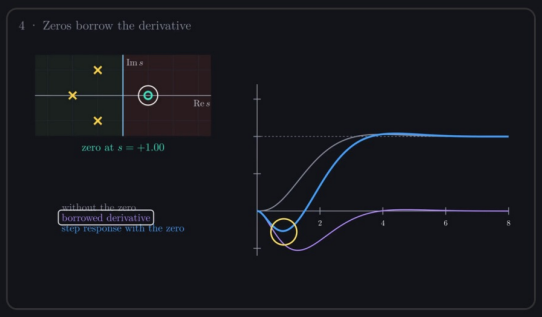


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Video 4 · Zeros borrow the derivative

Put a zero at z without changing G(0): $G = (s/(-z) + 1)\cdot\tilde{G} \Longrightarrow y = \tilde{y} + (1/(-z))\cdot d\tilde{y}/dt$

Left half-plane: positive derivative → earlier, faster, overshoot. Right half-plane (non-minimum phase): negative derivative → starts the **wrong way**. The poles, and so the stability, never change.



- Multiplying by s means d/dt because $\frac{d}{dt} e^{st} = s \cdot e^{st} = G(s) \cdot e^{st}$
- Multiplying by $1/s$ means $\int dt$ because $\int e^{st} dt = 1/s \cdot e^{st} = G(s) \cdot e^{st}$

Initial Value Theorem :

$$\lim_{t \to 0} y(t) = \lim_{s \to \infty} s \cdot F(s)$$

Final Value Theorem :

$$\lim_{t \to \infty} y(t) = \lim_{s \to 0} s \cdot F(s)$$

$$F(s) = \begin{cases} G(s) & \text{for impulse} \\ \frac{G(s)}{s} & \text{for step} \end{cases}$$

Your Turn

Question 12 *Choose the correct answer. (1 Point)*

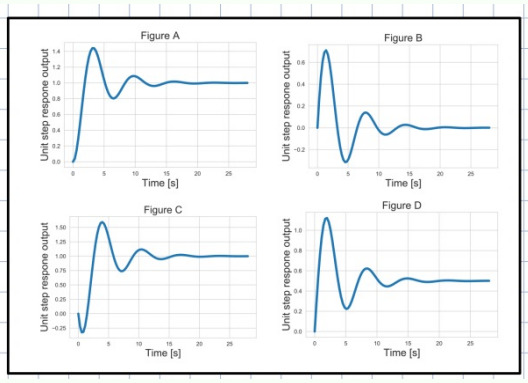
What are the initial y_0 and final values y_∞ of an *impulse response* for the following input to output transfer function?

$$G(s) = \frac{s+1}{2s^2 + 0.5s + 1}$$

A $y_0 = 0.5, \quad y_\infty = 0$
B $y_0 = 1, \quad y_\infty = 0$
C $y_0 = 0, \quad y_\infty = 1$
D $y_0 = 0, \quad y_\infty = 0.5$

You are given the following set of input to output transfer functions:

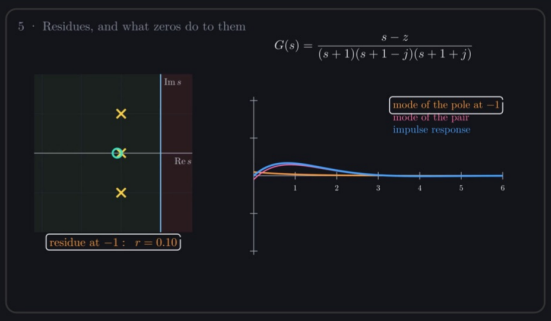
1. $G(s) = \frac{s+0.5}{s^2 + 0.5s + 1}$
2. $G(s) = \frac{1-s}{s^2 + 0.5s + 1}$
3. $G(s) = \frac{s}{s^2 + 0.5s + 1}$
4. $G(s) = \frac{1}{s^2 + 0.5s + 1}$



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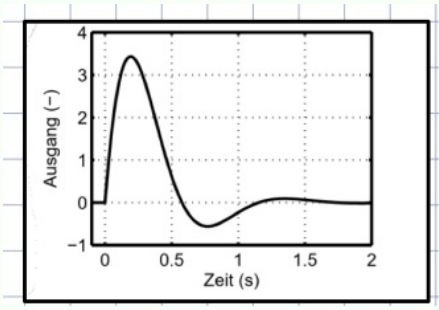
Video 5 · Residues, and what zeros do to them

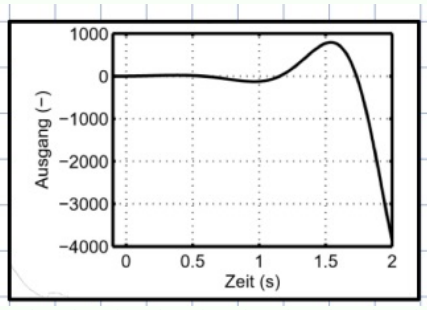
The residue is the same arrow picture, but **standing on the pole**: $r_i = k \cdot \Pi(p_i - z) / \Pi_{j \neq i}(p_i - p_j)$. A zero next to a pole makes that residue tiny: the mode is muffled. Exact cancellation hides the mode from G, but **not** from the system.

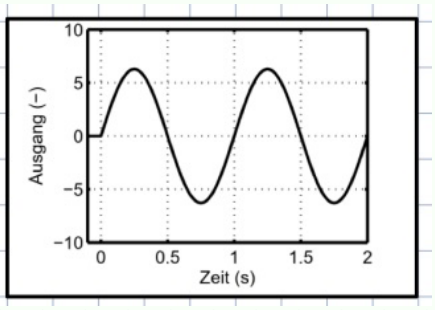


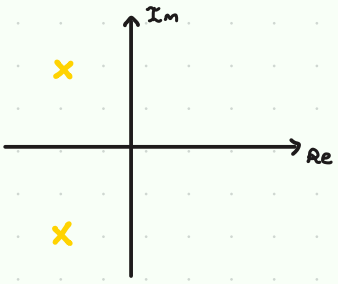
Made with Goodnotes

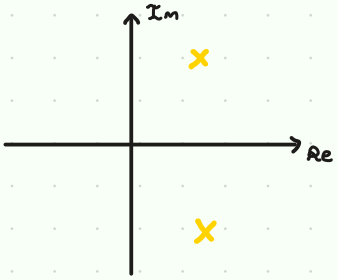
Your turn: Draw the poles of the systems by their impulse response

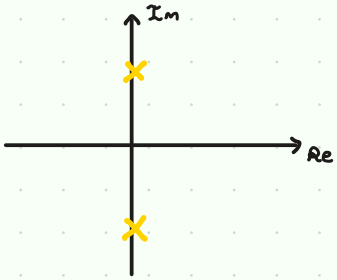












Your Turn

Question 12 *Choose the correct answer. (1 Point)*

What are the initial y_0 and final values y_∞ of an *impulse response* for the following input to output transfer function?

$$G(s) = \frac{s+1}{2s^2+0.5s+1}$$

■ $y_0 = 0.5, \quad y_\infty = 0$

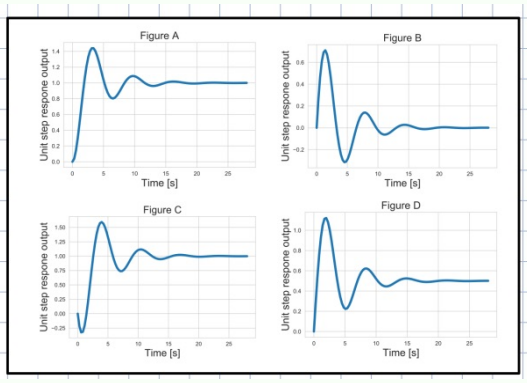
C $y_0 = 0, \quad y_\infty = 1$

B $y_0 = 1, \quad y_\infty = 0$

D $y_0 = 0, \quad y_\infty = 0.5$

You are given the following set of input to output transfer functions:

1. D
   $\lim_{t\to\infty} y(t) = \lim_{s\to 0} G(s) = 1/2$
   $G(s) = \frac{s+0.5}{s^2+0.5s+1}$
2. C
   undershoot
   $G(s) = \frac{1-s}{s^2+0.5s+1}$
3. B
   $\lim_{t\to\infty} y(t) = \lim_{s\to 0} G(s) = 0$
   $G(s) = \frac{s}{s^2+0.5s+1}$
4. A
   exclusion
   $G(s) = \frac{1}{s^2+0.5s+1}$



Made with Goodnotes